

ORIGINAL CONTRIBUTION

Sodium-Glucose Cotransporter 2 Inhibitor Use and Functional Outcomes After Endovascular Thrombectomy in Diabetic Stroke

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BACKGROUND: Although preclinical studies and limited clinical evidence suggest potential benefits of sodium-glucose cotransporter 2 inhibitors (SGLT2i) after acute ischemic stroke, their effects in patients with moderate-to-severe acute ischemic stroke requiring endovascular thrombectomy remain unclear. This study aimed to evaluate whether SGLT2i use after endovascular thrombectomy is associated with improved 3-month functional outcomes in diabetic patients with acute ischemic stroke.

METHODS: This retrospective cohort study analyzed prospectively collected data from January 2020 to March 2025 across 4 comprehensive stroke centers in Korea, Italy, and Switzerland. Consecutive adult patients with diabetes were grouped according to whether SGLT2i was used after their endovascular thrombectomy. Propensity score-based methods—including overlap weighting, inverse probability of treatment weighting, and 1:1 propensity score matching—were applied. Outcomes included the shift in 3-month modified Rankin Scale score, favorable (modified Rankin Scale score 0–2), and excellent outcomes (modified Rankin Scale score 0–1) at 3 months. Ordinal and binary logistic regression analyses were performed.

RESULTS: Of 429 eligible patients (mean [SD] age, 73.5 [11.1] years; 249 [58.0%] men; median [interquartile range] initial National Institutes of Health Stroke Scale score, 12 [6–18]), 51 (11.9%) received SGLT2i poststroke. Successful recanalization was achieved in 85.3%. In the overlap weighting model (weighted $n=37$ per group), SGLT2i use was associated with improved 3-month modified Rankin Scale score distribution (common odds ratio, 2.31 [95% CI, 1.30–4.10]) and higher odds of a favorable outcome (odds ratio, 2.26 [95% CI, 1.15–4.43]). Results were consistent across inverse probability of treatment weighting and propensity score matching models, with inverse probability of treatment weighting analyses additionally demonstrating significant associations with excellent outcomes. No significant effect modification was observed across key clinical subgroups.

CONCLUSIONS: Poststroke SGLT2i therapy after endovascular thrombectomy was associated with improved 3-month functional outcomes in this multicenter retrospective analysis of diabetic patients. These findings suggest a potential therapeutic role for SGLT2i in poststroke diabetes management and support future prospective trials.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: heart failure ■ glucose ■ ischemic stroke ■ sodium ■ thrombectomy

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) have demonstrated substantial cardiovascular benefits beyond glucose lowering in large-scale clinical trials.^{1–3} Reflecting this evidence, contemporary diabetes

guidelines recommend SGLT2i—along with glucagon-like peptide-1 receptor agonists—as preferred therapy for patients with type 2 diabetes who have atherosclerotic cardiovascular disease or elevated cardiovascular

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
CKD	chronic kidney disease
cOR	common odds ratio
EVT	endovascular thrombectomy
HF	heart failure
IPTW	inverse probability of treatment weighting
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
OR	odds ratio
OW	overlap weighting
PSM	propensity score matching
SGLT2i	sodium-glucose cotransporter 2 inhibitor

risk.⁴ As a result, SGLT2i have become central to cardiovascular protection in high-risk diabetic populations.

Despite these established benefits, the role of SGLT2i in cerebrovascular disease remains less certain; their effect on ischemic stroke incidence appeared rather modest.^{1–3} Early concerns also suggested that their glucosuric and diuretic properties could cause dehydration⁵ and potentially exacerbate stroke risk. However, subsequent meta-analyses and post hoc evaluations have shown no increased risk of stroke among SGLT2i users,^{6,7} and recent preclinical studies have highlighted possible neuroprotective mechanisms by reducing infarct volume in models of ischemic stroke.^{8–12} These findings provide a biological rationale for exploring the beneficial impact of SGLT2i on stroke outcomes.

We previously reported that among diabetic patients with acute ischemic stroke (AIS), SGLT2i use was not associated with in-hospital neurological deterioration, but was linked to improved functional outcomes at 3 months.¹³ However, that study primarily included patients with minor strokes and included only 1 center, limiting its generalizability.

To address these limitations, the present study aimed to evaluate diabetic patients with moderate-to-severe AIS undergoing endovascular thrombectomy (EVT) using data from both Asian and European stroke centers. We hypothesized that SGLT2i use after EVT would be associated with improved medium-term functional outcomes in this more clinically severe and ethnically diverse population.

METHODS

Data Availability

The data supporting the findings of this study are available from the corresponding author on reasonable request.

Study Population

Prospectively collected registries from 4 comprehensive stroke centers—Asan Medical Center and Seoul National University Hospital (Seoul, Korea); Maggiore Hospital (Bologna, Italy); and Lausanne University Hospital (Lausanne, Switzerland)—were used. Consecutive patients presenting with AIS between January 2020 and March 2025 who underwent EVT and had preexisting or newly diagnosed diabetes were included and grouped according to prescription or not of an SGLT2i at hospital discharge. Patients who died before discharge or were lost to follow-up at 3 months were excluded. The study was approved by the Institutional Review Boards of Asan Medical Center (No. 2025-0936) and Seoul National University Hospital (No. 1009-062-332), which waived the requirement for informed consent because of the retrospective study design. For the Bologna cohort, the use of anonymized data was approved by the local authority (IRCCS Istituto delle Scienze Neurologiche, Bologna, Italy; DL 196/2003). For the Lausanne cohort, as this study constituted a quality assurance project using anonymized data, ethical commission approval and patient consent were not required under the Swiss Human Research Act. This study was reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines, with the completed checklist provided in the [Supplemental Material](#).

Data Collection

Demographic information and vascular risk factors included age, sex, hypertension, hyperlipidemia, smoking status, prior stroke, atrial fibrillation, coronary heart disease, heart failure (HF), chronic kidney disease (CKD), and active cancer. HF was defined as a known diagnosis or a left ventricular ejection fraction $\leq 40\%$. CKD was defined as a documented diagnosis or estimated glomerular filtration rate ≤ 60 mL/(min $\cdot$ 1.73 m²), excluding cases of acute kidney injury that recovered.¹⁴ Active cancer included malignancies under current treatment, treated within the preceding 6 months, or under cancer-related palliative care. Medication history included prior antithrombotic and statin therapy and preadmission and discharge antidiabetic medications, including SGLT2i. Stroke-related characteristics comprised prestroke modified Rankin Scale (mRS) score, initial National Institutes of Health Stroke Scale (NIHSS) score, intravenous thrombolysis, occlusion site (anterior versus posterior circulation), onset-to-puncture time, successful recanalization, post-EVT NIHSS score, stroke cause, and symptomatic hemorrhagic transformation. Successful recanalization was defined as a modified Thrombolysis in Cerebral Infarction grade of 2b to 3. Post-EVT NIHSS score was measured between 12 and 36 hours after EVT puncture; when multiple assessments were available, the value closest to 24 hours was used. Stroke cause was determined according to the Trial of ORG 10172 in Acute Stroke Treatment classification and further grouped into 3 categories: large artery atherosclerosis, cardioembolism, and others. Symptomatic hemorrhagic transformation was defined as hemorrhage within the infarct area accompanied by a ≥ 4 -point increase in the NIHSS score. Baseline systolic blood pressure and laboratory parameters—including white blood cell count, hemoglobin, platelet count, low-density lipoprotein cholesterol, initial random glucose, and hemoglobin A1c—were also collected.



SGLT2i Use and Clinical Outcomes

SGLT2i use after thrombectomy was defined by prescription at discharge, comprising both continuation of prestroke therapy and de novo initiation after the index stroke. For patients who used SGLT2i, the time from stroke onset to the first SGLT2i administration during the index admission was recorded. The 3-month mRS score was assessed by stroke-trained physicians or research nurses at each participating center, either in person at an outpatient visit or by structured telephone interview when in-person assessment was not feasible. The primary outcome was the mRS score at 3 months. Favorable and excellent outcomes were defined as mRS score 0 to 2 and 0 to 1, respectively.

Statistical Analysis

Continuous variables were compared using the Student *t* test or Wilcoxon rank-sum test, and categorical variables using the χ^2 test or Fisher exact test. Temporal trends in SGLT2i prescription across study years were assessed using the Cochran–Armitage trend test.

To reduce confounding from baseline imbalances between patients with and without SGLT2i use, 3 propensity score–based approaches were applied: overlap weighting (OW), inverse probability of treatment weighting (IPTW), and propensity score matching (PSM). OW assigns greater weights to patients with a higher probability of receiving either treatment, emphasizing the region of covariate overlap and achieving exact balance on measured covariates.¹⁵ The weighted sample size was calculated as the sum of individual weights, which may differ from the unweighted patient count.

Propensity scores were estimated using multivariable logistic regression incorporating clinically relevant covariates: age, sex, hypertension, hyperlipidemia, smoking, previous stroke, atrial fibrillation, coronary heart disease, HF, CKD, active cancer, prior antithrombotic use, prior statin use, prior antidiabetic use, prestroke mRS score, initial NIHSS score, intravenous thrombolysis, occlusion site, onset-to-puncture time, successful recanalization, stroke cause, symptomatic hemorrhagic transformation, baseline systolic blood pressure, white blood cell count, hemoglobin, platelet count, low-density lipoprotein cholesterol, initial random glucose, and hemoglobin A1c. Missing values were imputed via multiple imputation using sequential regression under a missing-at-random assumption before conducting propensity score weighting and matching analyses.¹⁶ The results were obtained from an imputed data set after undergoing 10 iterative processes to achieve convergence. Weights for IPTW and OW were derived from the estimated propensity scores to create pseudopopulations with balanced covariate distributions.

PSM used 1:1 nearest-neighbor matching algorithm without replacement. Covariate balance was evaluated using absolute standardized differences, with values >0.1 indicating meaningful imbalance. Because OW achieved the best overall covariate balance and statistical efficiency, it served as the primary analytic model. After weighting or matching, baseline characteristics were compared using weighted Student *t* test or χ^2 test for weighted analyses and conditional logistic regression for the matched sample. Between-group comparisons of 3-month functional outcomes were performed using the weighted χ^2 test or the Cochran–Mantel–Haenszel shift

test for weighted analyses, and their corresponding tests for matched samples. Ordinal logistic regression assessed mRS score shifts, and binary logistic regression assessed favorable and excellent outcomes. Robust standard errors were used to account for clustering. In IPTW analyses, imbalanced covariates after weighting (absolute standardized difference >0.1) were included as adjustment terms in the regression models. To assess the potential impact of unmeasured confounding on the observed associations, *E* values were calculated for the point estimates and the lower confidence limits derived from the OW model.¹⁷

Subgroup analyses for mRS score shift and favorable outcome used OW model with multiplicative interaction terms across clinically relevant factors, including age (<65 or ≥ 65 years), sex, HF, CKD, active cancer, initial NIHSS score (<12 or ≥ 12), intravenous thrombolysis, occlusion site, onset-to-puncture time (<360 or ≥ 360 minutes), successful recanalization, stroke cause, hemoglobin A1c (<7.0 or $\geq 7.0\%$), and study region (Korea or Europe). To examine whether the association differed by the intensity of concomitant antidiabetic treatment, we tested for interaction across the number of non-SGLT2i antidiabetic medication classes at discharge, categorized as 0 to 1, 2, or ≥ 3 . Several sensitivity analyses were performed. To isolate the effect of de novo SGLT2i initiation, we repeated the primary analysis excluding patients receiving an SGLT2i before the stroke. To account for stroke severity after thrombectomy, we repeated the analysis with post-EVT NIHSS score replacing initial NIHSS score in the propensity score model.

All statistical tests were 2-sided, with $P < 0.05$ considered statistically significant. Analyses were performed with R (version 4.2.1; R Foundation for Statistical Computing, Vienna, Austria) and SAS (version 9.4; SAS Institute, Cary, NC).

RESULTS

A total of 429 patients with diabetes AIS who underwent EVT were included (Figure 1), with a mean (SD) age of 73.5 (11.1) years; 249 patients (58.0%) were men. Median [interquartile range] prestroke mRS score and initial NIHSS scores were 0 [0–1] and 12 [6–18], respectively. Successful recanalization was achieved in 366 patients (85.3%). Diabetes was previously diagnosed in 382 patients (89.0%) and newly diagnosed in 47 (11.0%; 2 [3.9%] in the SGLT2i group versus 45 [11.9%] in the control group). Nearly all patients had type 2 diabetes; only 3 (0.7%) had type 1 diabetes, all in the control group. Enrollment by center was as follows: Asan Medical Center ($n=115$), Seoul National University Hospital ($n=50$), Ospedale Maggiore di Bologna ($n=155$), and center Hospitalier Universitaire Vaudois ($n=109$). Overall, 51 patients (11.9%) received SGLT2i at discharge (empagliflozin, $n=30$; dapagliflozin, $n=20$; and canagliflozin, $n=1$). Of these, 32 (62.7%) were continued on prestroke SGLT2i therapy and 19 (37.3%) were started de novo after the index stroke. Compared with the de novo group, the continued group more often had CKD and prior antidiabetic use, while hemoglobin A1c was higher in the de novo group. Successful

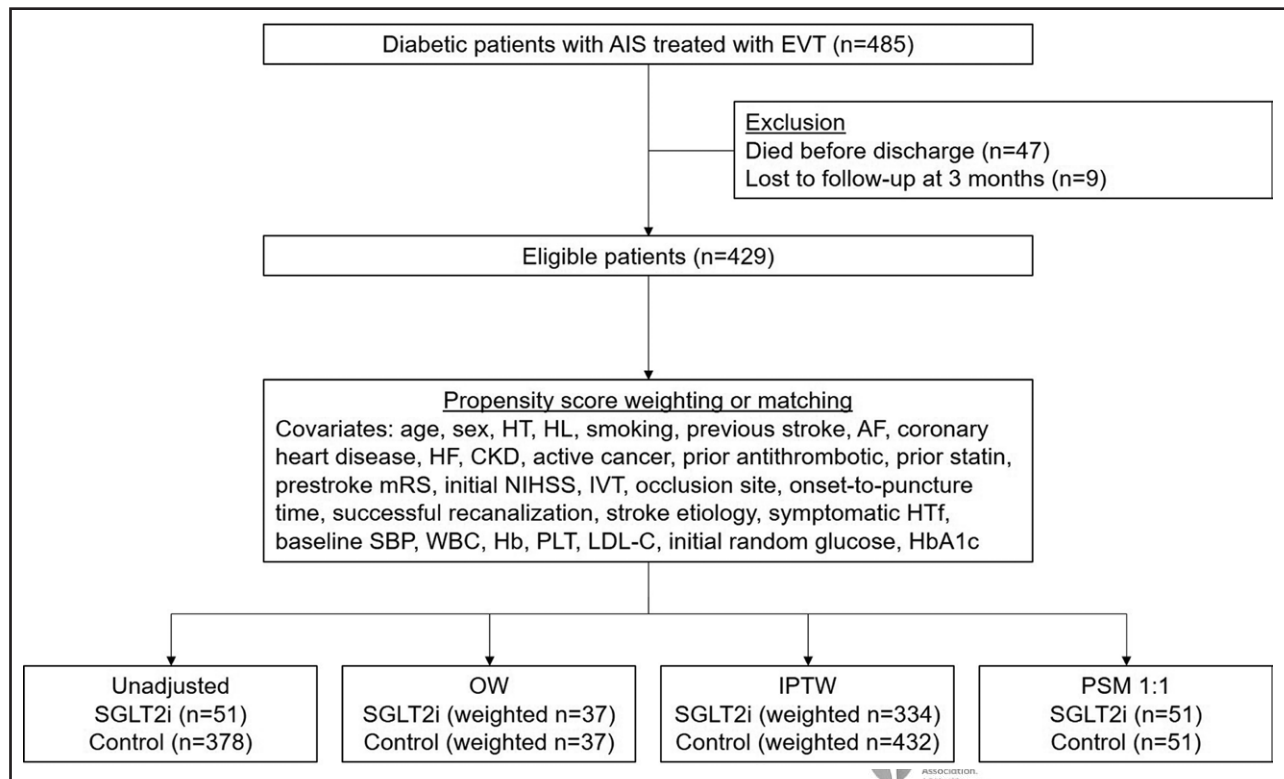


Figure 1. Flowchart.

AF indicates atrial fibrillation; AIS, acute ischemic stroke; CKD, chronic kidney disease; EVT, endovascular thrombectomy; Hb, hemoglobin; HbA1c, hemoglobin A1c; HF, heart failure; HL, hyperlipidemia; HT, hypertension; HTf, hemorrhagic transformation; IPTW, inverse probability of treatment weighting; IVT, intravenous thrombolysis; LDL-C, low-density lipoprotein cholesterol; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; OW, overlap weighting; PLT, platelet; PSM, propensity score matching; SBP, systolic blood pressure; SGLT2i, sodium-glucose cotransporter 2 inhibitor; and WBC, white blood cell.

recanalization was more frequent in the continued group, but 3-month outcomes were comparable (Table S1). The median [interquartile range] time from stroke onset to the first SGLT2i administration was 56.0 [32.6–104.0] hours. The proportion of SGLT2i prescriptions increased steadily over time, from 4.5% in 2020 to 21.1% in 2024 to 2025 (P for trend <0.001; Figure S1).

Before adjustment, patients in the SGLT2i group more frequently had a history of smoking, HF, and prior anti-thrombotic use. They also had shorter onset-to-puncture times, higher hemoglobin levels, lower low-density lipoprotein cholesterol levels, and lower initial blood pressure compared with patients not prescribed SGLT2i. Use of other antidiabetic medications is summarized in Table S2. Missing data were primarily observed in laboratory variables (2.3–11.0%), prior statin use (14.9%), and symptomatic hemorrhagic transformation (1.4%). Regional differences were also evident. Korean patients more frequently had previous stroke, active cancer, and large artery atherosclerosis, whereas CKD and prior antidiabetic treatment were more common in European patients. Intravenous thrombolysis was performed more often in Europe, but onset-to-puncture times were shorter in Korea. SGLT2i use was also more prevalent in Korea (Table S3).

Propensity score analyses produced the following weighted and matched group: OW produced weighted sample sizes of 37 per group with perfect covariate balance; IPTW resulted in weighted sample sizes of 334 in the SGLT2i group and 432 in the control group, and PSM yielded 51 matched pairs. In contrast to the OW model, some covariates remained imbalanced after IPTW or PSM (Figure S2). Baseline characteristics before and after applying propensity score-based adjustments are summarized in Table 1 and Table S4.

Patients treated with SGLT2i had better 3-month mRS scores compared with controls (SGLT2i versus control, median [interquartile range], 2 [1–3] versus 3 [2–4]; P <0.001). This difference remained significant across all analytic models: OW (2 [1–3] versus 3 [2–4]; P =0.047), IPTW (2 [1–3] versus 3 [2–4]; P <0.001), and PSM (2 [1–3] versus 3 [2–4]; P =0.005). In the OW model, SGLT2i use was associated with a favorable shift in the 3-month mRS score (common odds ratio [cOR], 2.31; 95% CI, 1.30–4.10; P =0.004) and higher odds of favorable outcomes (OR, 2.26 [95% CI, 1.15–4.43]; P =0.018). The association with excellent outcomes did not reach statistical significance. The corresponding E value (95% lower confidence limit) was 4.05 (1.92) for the mRS score shift and 3.95 (1.56)

Table 1. Baseline Characteristics Before and After Overlap Weighting

Variables	Before propensity score adjustment			After overlap weighting		
	SGLT2i (n=51)	Control (n=378)	P value	SGLT2i (weighted n=37)	Control (weighted n=37)	P value
Age, y	71.8±8.5	73.7±11.3	0.15	72.0±6.9	72.0±3.6	>0.99
Male sex	33 (64.7)	216 (57.1)	0.30	24 (65.4)	24 (65.4)	>0.99
Hypertension	42 (82.4)	302 (79.9)	0.68	30 (81.1)	30 (81.1)	>0.99
Hyperlipidemia	35 (68.6)	225 (59.5)	0.21	25 (68.3)	25 (68.3)	>0.99
Ever smoking	24 (47.1)	113 (29.9)	0.014	17 (45.1)	17 (45.1)	>0.99
Previous stroke	18 (35.3)	88 (23.3)	0.062	13 (34.4)	13 (34.4)	>0.99
Atrial fibrillation	26 (51.0)	155 (41.0)	0.18	18 (48.3)	18 (48.3)	>0.99
Coronary heart disease	10 (19.6)	74 (19.6)	>0.99	7 (20.1)	7 (20.1)	>0.99
Heart failure	14 (27.5)	45 (11.9)	0.003	9 (23.4)	9 (23.4)	>0.99
Chronic kidney disease	15 (29.4)	135 (35.7)	0.38	11 (29.1)	11 (29.1)	>0.99
Active cancer	3 (5.9)	44 (11.6)	0.22	2 (6.7)	2 (6.7)	>0.99
Prior antithrombotic medication	33 (64.7)	187 (49.5)	0.041	23 (62.3)	23 (62.3)	>0.99
Prior statin	33 (64.7)	195 (51.6)	0.078	23 (63.7)	23 (63.7)	>0.99
Prior antidiabetic medication	41 (80.4)	280 (74.1)	0.33	29 (79.8)	29 (79.8)	>0.99
Prestroke mRS score of 0–1	36 (70.6)	295 (78.0)	0.23	27 (73.4)	27 (73.4)	>0.99
Initial NIHSS score	10 [5–16]	12 [7–18]	0.30	10 [5–16]	11 [6–17]	>0.99
Intravenous thrombolysis	16 (31.4)	141 (37.3)	0.41	12 (33.7)	12 (33.7)	>0.99
Occlusion site			0.36			>0.99
Anterior circulation	46 (90.2)	323 (85.4)		32 (88.0)	32 (88.0)	
Posterior circulation	5 (9.8)	55 (14.6)		4 (12.0)	4 (12.0)	
Onset-to-puncture time, min	357 [190–815]	684 [263–867]	0.026	360 [206–860]	498 [178–787]	>0.99
Successful recanalization	46 (90.2)	320 (84.7)	0.29	33 (90.6)	33 (90.6)	>0.99
Stroke cause			0.71			>0.99
LAA	14 (27.5)	97 (25.7)		10 (27.0)	10 (27.0)	
CE	25 (49.0)	171 (45.2)		18 (48.1)	18 (48.1)	
Others	12 (23.5)	110 (29.1)		9 (24.9)	9 (24.9)	
Symptomatic HTf	0 (0.0)	13 (3.4)	0.38	0 (0.0)	0 (0.0)	...
Initial SBP, mm Hg	145.7±22.6	155.8±30.0	0.005	146.6±19.4	146.6±8.1	>0.99
WBC, ×10 ³ /μL	9.2±3.0	9.6±3.6	0.50	9.2±2.6	9.2±0.9	>0.99
Hemoglobin, g/dL	13.6±2.1	13.0±2.3	0.046	13.5±1.8	13.5±0.7	>0.99
Platelet, ×10 ³ /μL	235.3±71.2	230.5±86.8	0.70	234.1±63.0	234.1±28.2	>0.99
LDL-cholesterol, mg/dL	73.4±40.2	94.8±40.0	<0.001	77.3±35.6	77.3±10.5	>0.99
Initial random glucose, mg/dL	160.2±52.9	177.0±79.6	0.051	161.7±45.3	161.7±18.4	>0.99
Hemoglobin A1c, %	7.29±1.28	7.05±1.50	0.28	7.28±1.09	7.28±0.52	>0.99

Data are presented as mean±SD, median [interquartile range], or n (%), as appropriate. CE indicates cardioembolism; HTf, hemorrhagic transformation; LAA, large artery atherosclerosis; LDL, low-density lipoprotein; mRS, modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; SBP, systolic blood pressure; SGLT2i, sodium-glucose cotransporter 2 inhibitor; and WBC, white blood cell.

for favorable outcomes. Using IPTW model, associations remained significant for the mRS score shift (cOR, 3.30; 95% CI, 1.70–6.40; $P<0.001$), favorable outcomes (OR, 2.87 [95% CI, 1.26–6.52]; $P=0.012$), and excellent outcomes (OR, 2.55 [95% CI, 1.12–5.84]; $P=0.026$). These relationships persisted after adjusting for covariates that remained imbalanced postweighting. PSM results also aligned with the primary findings, where SGLT2i use remained associated with better functional outcomes (mRS score shift: cOR, 2.78; 95% CI, 1.27–6.08; $P=0.011$; favorable outcome: OR, 2.68

[95% CI, 1.05–6.86]; $P=0.039$). Detailed outcome distributions and comparative analyses are shown in Figure 2 and Table 2.

Subgroup analyses using the OW model revealed no significant interactions between SGLT2i use and any clinically relevant factors, indicating that the beneficial association with 3-month functional outcomes was consistent across subgroups (Figure 3). This association was not significantly modified by the number of concomitant non-SGLT2i antidiabetic medications (Table S5). In sensitivity analyses, the favorable association

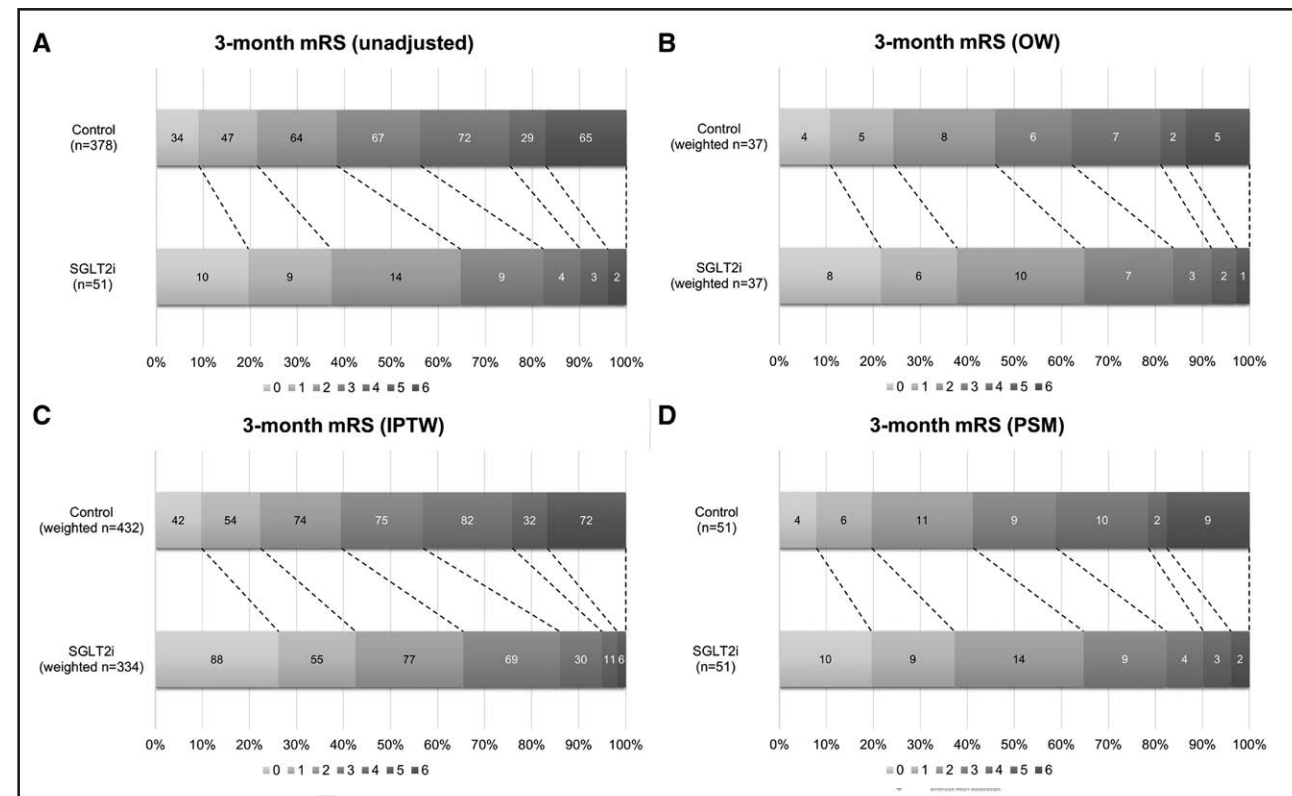


Figure 2. Functional outcomes at 3 months before and after propensity score weighting or matching.

Distribution of 3-month modified Rankin Scale (mRS) scores according to sodium-glucose cotransporter 2 inhibitor (SGLT2i) use: (A) before propensity score adjustment, (B) after overlap weighting (OW), (C) after inverse probability of treatment weighting (IPTW), and (D) after propensity score matching (PSM).

with the mRS score shift was preserved when the analysis was restricted to patients with no prior SGLT2i use (cOR, 2.40 [95% CI, 1.00–5.75]; $P=0.049$; Tables S6 and S7). When the post-EVT NIHSS score replaced the initial NIHSS score in the propensity score model, the association with the 3-month mRS score shift was directionally consistent across models, although statistical significance varied: borderline with OW (cOR, 1.76 [95% CI, 0.99–3.15]; $P=0.055$), significant with IPTW (cOR, 2.52 [95% CI, 1.35–4.70]; $P=0.004$), and nonsignificant with PSM (cOR, 1.60 [95% CI, 0.70–3.68]; $P=0.27$). It remained significant after directly adjusting for post-EVT NIHSS score in the full cohort (adjusted cOR, 1.90 [95% CI, 1.06–3.41]; $P=0.031$; Tables S8 and S9).

DISCUSSION

In this multicenter cohort of diabetic patients with moderate-to-severe AIS treated with EVT, SGLT2i use was associated with a favorable shift in the 3-month mRS scores and higher odds of achieving favorable outcomes. These associations were robust across multiple propensity score approaches and remained consistent in clinically relevant subgroups, suggesting that the favorable associations with SGLT2i may be broadly generalizable to diverse populations and healthcare settings. The

steady increase in SGLT2i use during the study period further underscores their growing clinical adoption and the relevance of evaluating their role in stroke recovery.

Efforts to improve poststroke functional outcomes have historically yielded inconclusive or disappointing results.^{18,19} Notably, a recent randomized controlled trial demonstrated that liraglutide improved functional outcomes at 3 months in patients with minor ischemic stroke or transient ischemic attack,²⁰ renewing interest in the neurological effects of cardiometabolic therapies. However, despite belonging to the same broad class of cardiometabolic agents, evidence supporting a similar benefit of SGLT2i remains limited. Few studies have assessed their role in poststroke functional recovery, and none has specifically focused on moderate-to-severe stroke or EVT-treated patients—groups at high risk for poor outcomes due to larger infarct burdens and reperfusion stress. Our findings extend current evidence by demonstrating that potential benefits of SGLT2i are not restricted to minor strokes, as shown in our previous study,¹³ but may also apply to more severe neurological injury. These results support the hypothesis that SGLT2i promote recovery through mechanisms beyond glucose control.

Although our study cannot determine causality, several biological mechanisms may explain the observed

Table 2. Association of SGLT2i Use With 3-Month Functional Outcomes

	SGLT2i	Control	P value	OR/cOR (95% CI)*	P value	Adjusted OR/cOR (95% CI)*	P value
Unadjusted	n=51	n=378					
3-mo mRS score	2 [1–3]	3 [2–4]	<0.001	2.72 (1.62–4.59)	<0.001	2.52 (1.42–4.48)†	0.002
Favorable outcome	33 (64.7)	145 (38.4)	<0.001	2.95 (1.60–5.43)	<0.001	2.81 (1.26–6.31)†	0.012
Excellent outcome	19 (37.3)	81 (21.4)	0.012	2.18 (1.17–4.04)	0.014	2.36 (1.06–5.26)†	0.036
OW	Weighted n=37	Weighted n=37					
3-mo mRS score	2 [1–3]	3 [2–4]	0.047	2.31 (1.30–4.10)	0.004	...	...
Favorable outcome	24 (65.1)	17 (45.3)	0.017	2.26 (1.15–4.43)	0.018	...	...
Excellent outcome	14 (38.2)	9 (24.7)	0.077	1.88 (0.93–3.80)	0.079	...	...
IPTW	Weighted n=334	Weighted n=432					
3-mo mRS score	2 [1–3]	3 [2–4]	<0.001	3.30 (1.70–6.40)	<0.001	3.62 (2.05–6.37)‡	<0.001
Favorable outcome	219 (65.3)	171 (39.6)	0.008	2.87 (1.26–6.52)	0.012	3.70 (1.78–7.69)‡	<0.001
Excellent outcome	143 (42.4)	97 (22.4)	0.018	2.55 (1.12–5.84)	0.026	3.14 (1.47–6.72)‡	0.003
PSM	n=51	n=51					
3-mo mRS score	2 [1–3]	3 [2–4]	0.005	2.78 (1.27–6.08)	0.011	...	...
Favorable outcome	33 (64.7)	21 (41.2)	0.014	2.68 (1.05–6.86)	0.039	...	...
Excellent outcome	19 (37.3)	10 (19.6)	0.051	2.38 (0.90–6.29)	0.080	...	...

Data are presented as median [interquartile range] or n (%), as appropriate. cOR indicates common odds ratio; IPTW, inverse probability of treatment weighting; mRS, modified Rankin Scale; OR, odds ratio; OW, overlap weighting; PSM, propensity score matching; and SGLT2i, sodium-glucose cotransporter 2 inhibitor.

*ORs were estimated from logistic regression for binary outcomes, and cORs from ordinal logistic regression for the 3-mo mRS score shift, calculated for lower mRS scores.

†Adjusted for variables included in the propensity score model.

‡Adjusted for variables with absolute standardized difference >0.1 after IPTW, including age, sex, smoking, previous stroke, chronic kidney disease, prior statin, prior antidiabetic medication, prestroke mRS score, successful recanalization, stroke cause, and hemoglobin. Symptomatic hemorrhagic transformation also had an absolute standardized difference >0.1 but was not included because it did not occur in the SGLT2i group.

association. In experimental models of cerebral ischemia–reperfusion, SGLT2 inhibition markedly reduced infarct volume independent of glycemic control,^{8–12} suggesting an early neuroprotective effect against acute ischemic injury. In addition, SGLT2i have been shown to preserve pericyte–blood–brain barrier integrity, supporting neurovascular remodeling and recovery.^{8,21–23} Notably, SGLT2 inhibition reduced infarct volume by limiting pericyte loss within the ischemic territory and enhancing mitochondrial biogenesis, which increases tissue resistance to ischemic stress.⁸ In another study, post-stroke empagliflozin treatment normalized diabetes-induced aberrant pericyte distribution and improved motor recovery without a concomitant reduction in infarct size, indicating that SGLT2i may promote functional recovery independent of initial lesion reduction.²¹ Furthermore, SGLT2 inhibition may attenuate ischemia-induced activation of sodium-glucose cotransporter 1—more abundantly expressed in the brain than SGLT2 and implicated in neuronal apoptosis and neuroglial injury—thereby facilitating poststroke synaptic remodeling.^{12,24} Beyond neuroprotection, SGLT2 inhibition has shown the ability to reduce fat mass, upregulate genes related to browning of white adipose tissue, and improve oxidative stress.²⁵ Our previous study demonstrated increased metabolic activity in supraclavicular and subcutaneous adipose tissue among stroke survivors.¹³ Collectively, these pleiotropic actions, alongside

the established cardiometabolic benefits of SGLT2i, provide a biologically plausible explanation for the improved long-term functional recovery observed in moderate-to-severe AIS.

Limitations

Some limitations of our study should be acknowledged. First, despite including 4 international centers, the sample size remained modest, reflecting the relatively low prescription rates of SGLT2i among patients with diabetic stroke. This was particularly evident in the sensitivity analysis using post-EVT NIHSS score, where the association attenuated under OW and PSM. Because the post-EVT NIHSS score was lower in the SGLT2i group, adjustment shifted the population toward lower neurological severity, narrowing the between-group difference in outcomes, which our modest sample size was likely underpowered to detect. However, as observed in our cohort, the marked increase in SGLT2i use in recent years suggests that larger contemporary data will be available to validate these findings. Second, as a registry of EVT-treated patients, our cohort could not capture patients receiving an SGLT2i who did not undergo EVT, nor the reasons for nontreatment. The possibility of preprocedural selection, therefore, cannot be excluded, and its direction and magnitude cannot be determined. Third, although multiple imputation was used, missing data may

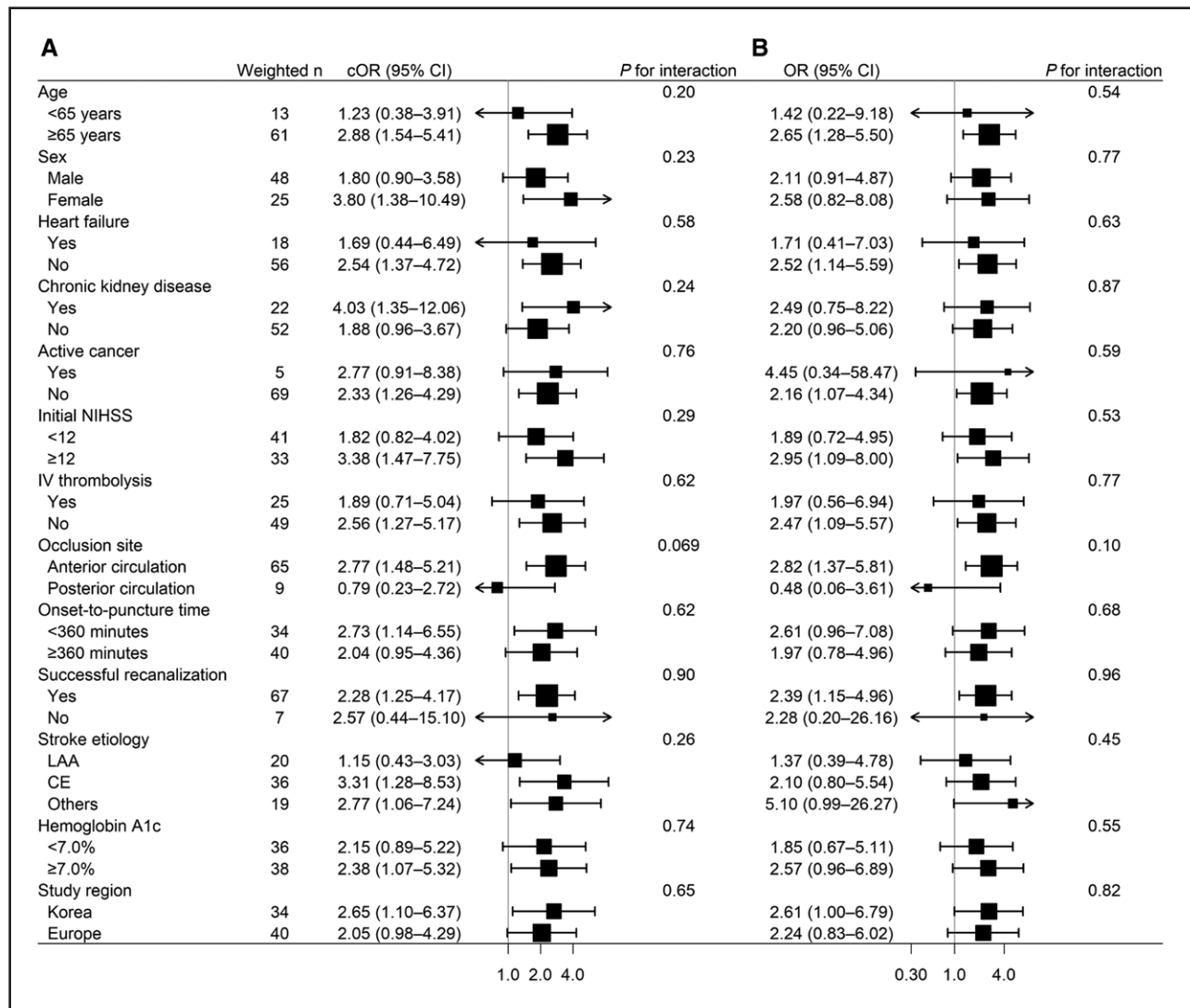


Figure 3. Subgroup analysis of functional outcomes at 3 months.

Forest plot presenting the association between sodium-glucose cotransporter 2 inhibitor (SGLT2i) use and 3-month functional outcomes, stratified by key clinical factors. **A**, Common odds ratios (cORs) for improvement in the 3-month modified Rankin Scale (mRS) score distribution and **B** odds ratios (ORs) for favorable outcomes were estimated using ordinal and binary logistic regression models, respectively, incorporating multiplicative interaction terms on overlap-weighted cohorts. The size of each square represents the relative weight of the subgroup in the analysis, while the error bars indicate 95% CIs. All estimates are displayed on a logarithmic scale along the x axis. CE indicates cardioembolism; IV, intravenous; LAA, large artery atherosclerosis; and NIHSS, National Institutes of Health Stroke Scale.

still introduce uncertainty. Fourth, unmeasured confounding—such as infarct volume or collateral status—cannot be fully excluded, though the use of multiple complementary propensity score-based methods likely minimized its impact on the results. The relatively large *E* values indicated that only an unmeasured confounder moderately to strongly associated with both SGLT2i use and functional outcomes could negate the observed association. That is, the likelihood of there being unmeasured confounding variables seems low. Relatedly, although the post-EVT NIHSS score may more directly reflect stroke severity at the time of SGLT2i initiation or continuation, it is not routinely registered and is assessed during an early, fluctuating period. Thus, a stable measure was unavailable,

and it was examined only in a sensitivity analysis. Finally, the retrospective design precludes causal inference; prospective studies specifically designed for stroke populations are warranted to validate these findings and elucidate the underlying mechanisms.

Conclusions

In this multicenter retrospective cohort study, SGLT2i therapy was associated with better 3-month functional outcomes in diabetic patients with moderate-to-severe AIS who underwent EVT. These findings highlight the potential value of integrating SGLT2i into poststroke management strategies and align with evolving diabetes

guidelines emphasizing cardiovascular protection in high-risk populations.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S9
Figures S1–S2
STROBE Checklist

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